

OS PRACTICAL

PRACTICAL-5

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir practical5
rishitha@Rishitha:~$ cd practical5
rishitha@Rishitha:~/practical5$ pwd
/home/rishitha/practical5
rishitha@Rishitha:~/practical5$ nano ProducerConsumer.c
rishitha@Rishitha:~/practical5$ gcc ProducerConsumer.c -o ProducerConsumer
rishitha@Rishitha:~/practical5$ ./ProducerConsumer
Parent Produced: Hello Child Process
Child Consumed: Hello Child Process
```

```
rishitha@Rishitha:~/practical5$ nano PipeExec.c
rishitha@Rishitha:~/practical5$ gcc PipeExec.c -o PipeExec
rishitha@Rishitha:~/practical5$ ./PipeExec
-rwxr-xr-x 1 rishitha rishitha 16256 Aug 17 05:51 PipeExec
-rw-r--r-- 1 rishitha rishitha  551 Aug 17 05:51 PipeExec.c
-rwxr-xr-x 1 rishitha rishitha 16304 Aug 17 05:50 ProducerConsumer
-rw-r--r-- 1 rishitha rishitha  616 Aug 17 05:49 ProducerConsumer.c
rishitha@Rishitha:~/practical5$
```